

UnHerd Majority

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```
##install.packages("tidyverse")
##install.packages("tidytext")
##install.packages("textdata")
##install.packages("rmarkdown")
##install.packages("ggplot2")
##install.packages("knitr")

library(tidyverse)

## Warning: package 'tidyverse' was built under R version 4.3.3
## Warning: package 'ggplot2' was built under R version 4.3.3
## Warning: package 'tidyr' was built under R version 4.3.3
## Warning: package 'readr' was built under R version 4.3.3

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2     3.5.1      v tibble     3.2.1
## v lubridate  1.9.3      v tidyr      1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(tidytext)

## Warning: package 'tidytext' was built under R version 4.3.3

library(textdata)

## Warning: package 'textdata' was built under R version 4.3.3

library(knitr)

## Warning: package 'knitr' was built under R version 4.3.3

library(rmarkdown)

## Warning: package 'rmarkdown' was built under R version 4.3.3

library(ggplot2)

# Load the data
data <- read_csv("Book1.csv")
```

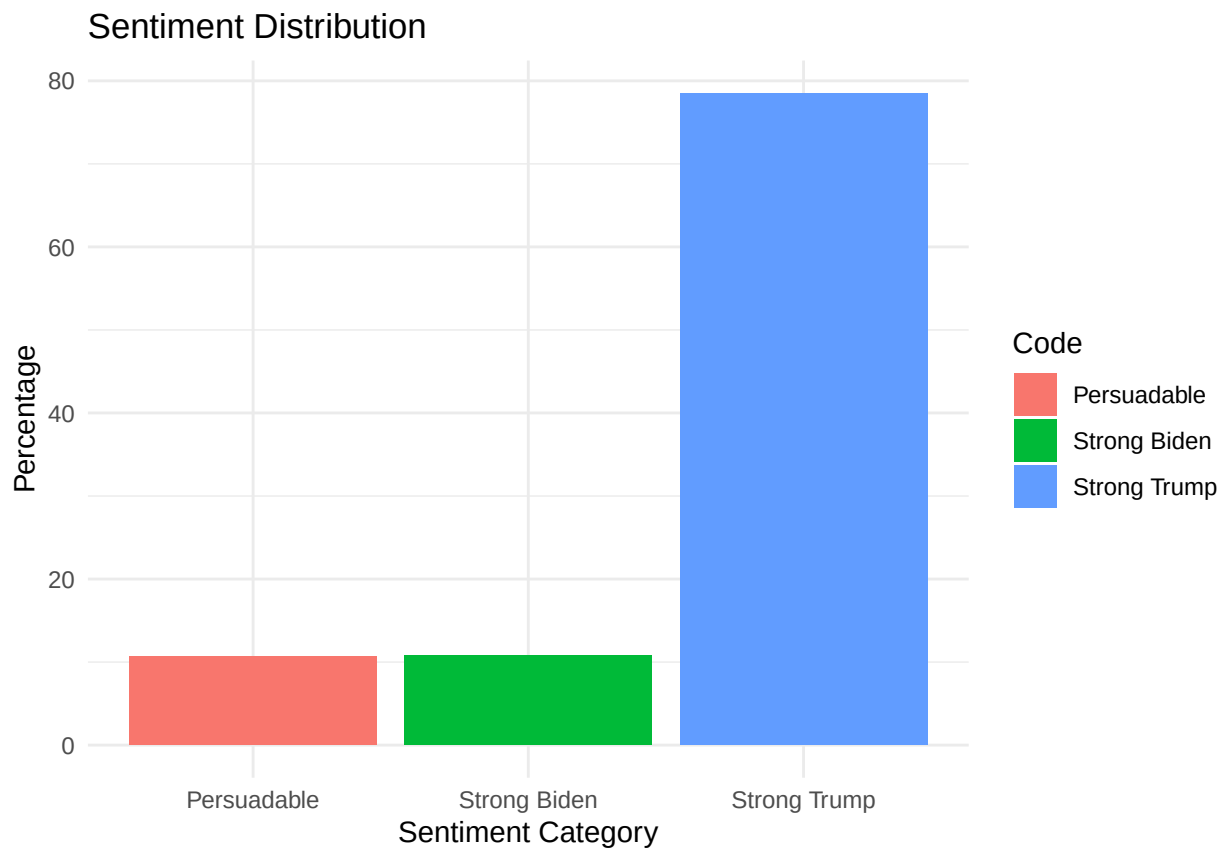
```
## Rows: 18215 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Audience, Ad Name, Created At, Commenter, Comment Message, Code
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Summary of Comments:

```
sentiment_distribution <- data %>%
  count(Code) %>%
  mutate(percentage = n / sum(n) * 100)
print(sentiment_distribution)
```

```
## # A tibble: 3 x 3
##   Code          n percentage
##   <chr>      <int>      <dbl>
## 1 Persuadable  1944      10.7
## 2 Strong Biden  1969      10.8
## 3 Strong Trump 14302     78.5
```

```
ggplot(sentiment_distribution, aes(x = Code, y = percentage, fill = Code)) +
  geom_bar(stat = "identity") +
  labs(title = "Sentiment Distribution", x = "Sentiment Category", y = "Percentage") +
  theme_minimal()
```



Content Eliciting Each Type of Comment

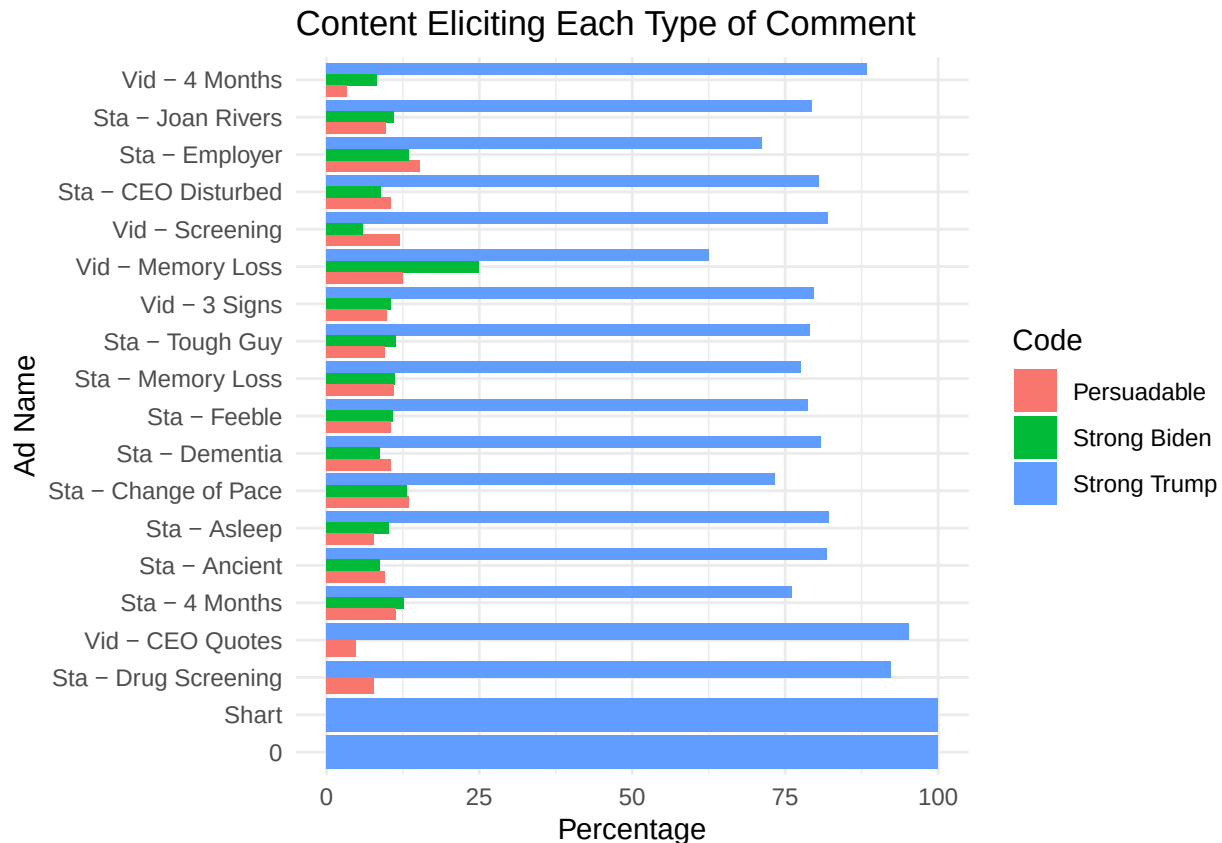
```
ad_summary <- data %>%
  group_by(`Ad Name`, Code) %>%
  summarize(count = n()) %>%
  mutate(total = sum(count)) %>%
  mutate(percentage = count / total * 100) %>%
  arrange(desc(percentage))
```

`summarise()` has grouped output by 'Ad Name'. You can override using the
`.groups` argument.

```
print(ad_summary)
```

```
## # A tibble: 51 x 5
## # Groups:   Ad Name [19]
##   `Ad Name`      Code      count total percentage
##   <chr>          <chr>    <int> <int>    <dbl>
## 1 0            Strong Trump    13    13      100
## 2 Shart        Strong Trump     1     1      100
## 3 Vid - CEO Quotes Strong Trump    20    21      95.2
## 4 Sta - Drug Screening Strong Trump    12    13      92.3
## 5 Vid - 4 Months Strong Trump    53    60      88.3
## 6 Sta - Asleep Strong Trump  2459  2995      82.1
## 7 Vid - Screening Strong Trump    82   100       82
## 8 Sta - Ancient Strong Trump  1264  1545      81.8
## 9 Sta - Dementia Strong Trump  1264  1565      80.8
## 10 Sta - CEO Disturbed Strong Trump  1293  1606      80.5
## # i 41 more rows
```

```
ggplot(ad_summary, aes(x = reorder(`Ad Name`, -percentage), y = percentage, fill = Code)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Content Eliciting Each Type of Comment", x = "Ad Name", y = "Percentage") +
  theme_minimal() +
  coord_flip()
```



Differences in Engagement and Comments Between Audiences

```
audience_summary <- data %>%
  group_by(Audience, Code) %>%
  summarize(count = n()) %>%
  mutate(total = sum(count)) %>%
  mutate(percentage = count / total * 100) %>%
  arrange(Audience, desc(percentage))
```

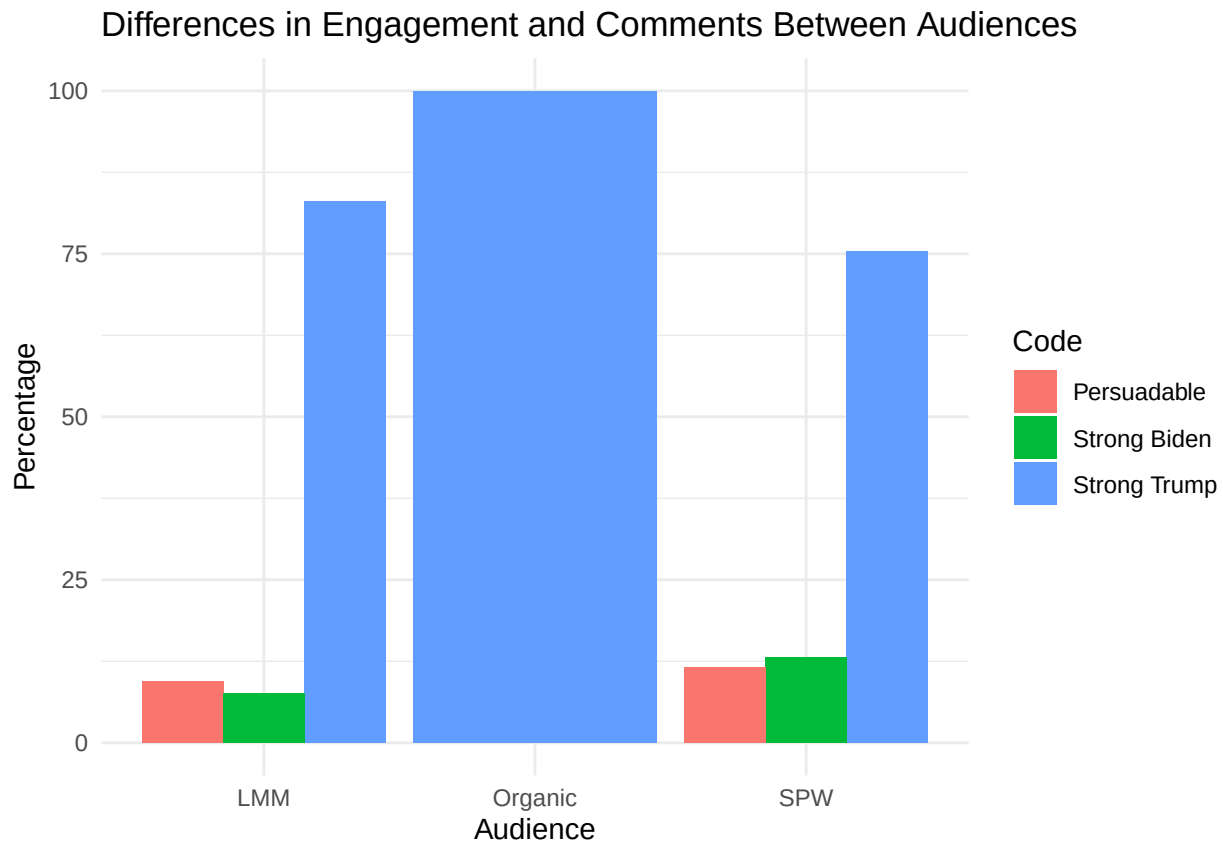
`summarise()` has grouped output by 'Audience'. You can override using the
`.groups` argument.

```
print(audience_summary)
```

```
## # A tibble: 7 x 5
## # Groups:   Audience [3]
##   Audience Code      count total percentage
##   <chr>      <chr>      <int> <int>      <dbl>
## 1 LMM      Strong Trump    6197  7464      83.0
## 2 LMM      Persuadable     701  7464       9.39
## 3 LMM      Strong Biden    566  7464       7.58
## 4 Organic Strong Trump     16   16      100
## 5 SPW      Strong Trump    8089 10735      75.4
## 6 SPW      Strong Biden    1403 10735      13.1
## 7 SPW      Persuadable    1243 10735      11.6
```

```
ggplot(audience_summary, aes(x = Audience, y = percentage, fill = Code)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Differences in Engagement and Comments Between Audiences", x = "Audience", y = "Percentage")
```

```
theme_minimal()
```



Policy Areas Resonance

```
policy_keywords <- list(
  Economy = c("economy", "economic", "finance", "financial", "market"),
  Jobs = c("job", "jobs", "employment", "unemployment", "work"),
  Taxes = c("tax", "taxes", "taxation", "IRS", "income tax"),
  Healthcare = c("healthcare", "health care", "insurance", "medicare", "medicaid")
)

tag_policy_area <- function(comment, keywords) {
  policies <- unlist(lapply(names(keywords), function(policy) {
    if (any(sapply(keywords[[policy]], function(keyword) {
      grepl(keyword, comment, ignore.case = TRUE)
    }))) {
      return(policy)
    }
  }))
  if (length(policies) == 0) {
    return("Other")
  } else {
    return(paste(policies, collapse = ", "))
  }
}

data <- data %>%
```

```
mutate(PolicyArea = sapply(`Comment Message`, tag_policy_area, keywords = policy_keywords))
head(data)
```

```
## # A tibble: 6 x 7
##   Audience `Ad Name`   `Created At` Commenter `Comment Message` Code PolicyArea
##   <chr>    <chr>        <chr>      <chr>    <chr>          <chr> <chr>
## 1 SPW      Vid - 3 Si~ 2024-06-12 ~ Michele ~ No way!!      Stro~ Other
## 2 LMM      Vid - 3 Si~ 2024-06-12 ~ John Roo~ All false     Stro~ Other
## 3 LMM      Vid - 3 Si~ 2024-06-12 ~ William ~ Wow all lies.. l~ Stro~ Other
## 4 LMM      Vid - 3 Si~ 2024-06-12 ~ Kiel Rei~ Cant wait to vot~ Stro~ Other
## 5 LMM      Vid - 3 Si~ 2024-06-12 ~ Andy Kru~ That man is stro~ Stro~ Other
## 6 SPW      Vid - 3 Si~ 2024-06-12 ~ Tina Sla~ you have the wro~ Pers~ Other
```

```
data <- data %>%
  mutate(PolicyArea = strsplit(PolicyArea, ", ")) %>%
  unnest(PolicyArea)
```

```
head(data)
```

```
## # A tibble: 6 x 7
##   Audience `Ad Name`   `Created At` Commenter `Comment Message` Code PolicyArea
##   <chr>    <chr>        <chr>      <chr>    <chr>          <chr> <chr>
## 1 SPW      Vid - 3 Si~ 2024-06-12 ~ Michele ~ No way!!      Stro~ Other
## 2 LMM      Vid - 3 Si~ 2024-06-12 ~ John Roo~ All false     Stro~ Other
## 3 LMM      Vid - 3 Si~ 2024-06-12 ~ William ~ Wow all lies.. l~ Stro~ Other
## 4 LMM      Vid - 3 Si~ 2024-06-12 ~ Kiel Rei~ Cant wait to vot~ Stro~ Other
## 5 LMM      Vid - 3 Si~ 2024-06-12 ~ Andy Kru~ That man is stro~ Stro~ Other
## 6 SPW      Vid - 3 Si~ 2024-06-12 ~ Tina Sla~ you have the wro~ Pers~ Other
```

```
sentiment_distribution <- data %>%
  count(Code) %>%
  mutate(percentage = n / sum(n) * 100)
```

```
ad_summary <- data %>%
  group_by(`Ad Name`, Code) %>%
  summarize(count = n(), .groups = 'drop') %>%
  mutate(total = sum(count)) %>%
  mutate(percentage = count / total * 100) %>%
  arrange(desc(percentage))
```

```
audience_summary <- data %>%
  group_by(Audience, Code) %>%
  summarize(count = n(), .groups = 'drop') %>%
  mutate(total = sum(count)) %>%
  mutate(percentage = count / total * 100) %>%
  arrange(Audience, desc(percentage))
```

```
policy_resonance <- data %>%
  filter(PolicyArea != "Other") %>%
  group_by(Audience, PolicyArea) %>%
  summarize(count = n(), .groups = 'drop') %>%
  mutate(total = sum(count)) %>%
  mutate(percentage = count / total * 100) %>%
  arrange(Audience, desc(percentage))
```

```

top_5_policy_resonance <- policy_resonance %>%
  group_by(Audience) %>%
  top_n(5, wt = percentage) %>%
  ungroup() %>%
  arrange(Audience, desc(percentage))

print(top_5_policy_resonance)

## # A tibble: 9 x 5
##   Audience PolicyArea count total percentage
##   <chr>      <chr>      <int> <int>      <dbl>
## 1 LMM       Jobs          235  1417      16.6
## 2 LMM       Taxes          197  1417      13.9
## 3 LMM       Economy         73  1417       5.15
## 4 LMM       Healthcare        8  1417      0.565
## 5 Organic   Taxes           3  1417      0.212
## 6 SPW       Jobs          422  1417      29.8
## 7 SPW       Taxes          283  1417      20.0
## 8 SPW       Economy        120  1417       8.47
## 9 SPW       Healthcare     76  1417       5.36

ggplot(top_5_policy_resonance, aes(x = reorder(PolicyArea, -percentage), y = percentage, fill = Audience)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Top 5 Policy Areas Resonance by Audience", x = "Policy Area", y = "Percentage") +
  theme_minimal() +
  coord_flip()

```

